GAN Attribution After DNS Fusion

Same real rows • same DNS model • same calibration cohort • same 20K URLs

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	FPR	FNR
Real only	96.73%	92.59%	73.15%	81.73%	97.67%	0.65%	26.85%
Real + DNS	97.02%	90.72%	78.20%	84.00%	97.70%	0.89%	21.80%
GAN +10%	97.55%	94.83%	79.85%	86.70%	97.52%	0.48%	20.15%
GAN +10% + DNS	97.69%	92.30%	83.90%	87.90%	97.56%	0.78%	16.10%

GAN effect with DNS: recall +5.70 points • F1 +3.90 • FPR -0.111 • FNR -5.70

With the same frozen DNS verifier, GAN augmentation improves seven of eight registered 20K metrics: recall by 5.70 points, F1 by 3.90, PR-AUC by 1.57, FPR by 0.111, and FNR by 5.70; ROC-AUC is 0.14 points lower. The GAN contribution therefore remains beneficial after DNS fusion under this controlled diagnostic protocol, but it does not improve every metric.

The DNS classifier contains zero synthesized URLs. GAN augmentation changes only the primary URL model. The protected 20K cohort is diagnostic because its labels were viewed previously.

Independent Development Holdout

3,290 registered-domain-disjoint rows • no threshold selection on this split

Condition	Precision	Recall	F1	ROC-AUC	FPR	FNR
Real only	98.54%	71.72%	83.02%	97.04%	0.43%	28.28%
Real + DNS	97.68%	75.95%	85.46%	97.06%	0.72%	24.05%
GAN +10%	98.52%	77.75%	86.92%	96.64%	0.47%	22.25%
GAN +10% + DNS	97.70%	80.93%	88.53%	96.66%	0.77%	19.07%

Holdout GAN effect with DNS: recall +4.98 points; F1 +3.07; FPR +0.043; FNR -4.98.

This split is the stronger internal check because the DNS threshold was chosen only on the separate calibration split. It remains same-snapshot rather than temporal evidence.

Controlled Experimental Contract

Component	Real condition	GAN condition	Control
Real training identities	367,184	367,184	Identical
Synthetic phishing rows	0	40,410	Only intended difference
Primary architecture	HYB-MS	HYB-MS	Identical
Primary seeds	42/43/44	42/43/44	Identical
Calibration cohort	23,100	23,100	Identical
Primary FPR rule	0.50% cap	0.50% cap	Identical rule
DNS training rows	607	607	Identical
DNS RF seeds	42/43/44	42/43/44	Identical
DNS threshold	0.504088	0.504088	Identical
20K URL identities	20,000	20,000	Identical

Each primary receives its own calibration-derived numerical threshold because their score scales differ. The cohort and threshold-selection rule are the same; forcing one raw threshold onto both would be methodologically invalid.

What the Comparison Proves

Result	With the same frozen DNS verifier, GAN augmentation improves seven of eight registered 20K metrics: recall by 5.70 points, F1 by 3.90, PR-AUC by 1.57, FPR by 0.111, and FNR by 5.70; ROC-AUC is 0.14 points lower. The GAN contribution therefore remains beneficial after DNS fusion under this controlled diagnostic protocol, but it does not improve every metric.
Attribution	Because the DNS branch is byte-identical, the combined-system difference is attributable to primary GAN augmentation plus its changed routing decisions.
No synthetic DNS	The generator produces URL text only. DNS training remains 607 real routed domains; no fabricated A/AAAA/CNAME/MX/NS values are used.
Operational caution	At low phishing prevalence, FPR dominates warning precision. DNS escalation should remain caution/review evidence rather than an automatic block.
Claim boundary	The 20K analysis is diagnostic and DNS is mutable. A future low-prevalence temporal cohort is still required for deployment confirmation.